

# Exposing the Invisible: Detecting Stealthy Parameter-Based Cyber-Attacks on Inverter Synchronization Loops

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**Abstract**—The increasing integration of Internet-of-Things (IoT) technologies for monitoring and control of inverter-based resources (IBRs) has expanded the attack surface, enabling stealthy manipulation of controller parameters through vulnerable supervisory control interfaces. Phase-locked loops (PLLs) emerge as prime targets, as they interact with all control loops and critically influence the dynamic response of grid-following (GFL) converters. This paper analyzes the underlying threat model to elucidate the mechanisms enabling such stealthy behavior and conducts a thorough stability and transient response analysis to characterize how PLL tampering can degrade system performance without necessarily destabilizing the system. The results reveal critical interactions among frequency estimation, control, and synchronization that can significantly reduce stability margins. To counter the stealthy nature of these attacks, a modified PLL is proposed that exposes gain variations through shifts in its equilibrium points while preserving conventional PLL performance. Experimental results validate the effectiveness of the proposed approach in detecting PLL cyber-attacks.

**Index Terms**—Phase-locked loop, cyber-attack, attack detection, stability analysis, synchronization.

## I. INTRODUCTION

The power grid is transitioning from centralized, generator-based operation to decentralized coordination of grid-connected distributed generation, including inverter-based-resources (IBRs) [1]. While this shift supports decarbonization, it introduces challenges such as reduced system inertia and damping, formation of weak or isolated networks prone to voltage disturbances and instability, and increased cybersecurity risks [2]–[4]. Poorly integrated Internet-of-Things (IoT) technologies in power plant coordination have expanded attack surfaces, allowing adversaries to exploit vulnerabilities in critical communication channels and potentially disrupt grid operations [5]–[7].

Supervisory control and data acquisition (SCADA)-based attacks are a credible threat, as evidenced by previous incidents. In particular, the cyber-attack reported in [8] demonstrates how adversaries can disrupt communication between utilities and IBRs in California and Wyoming, exposing critical vulnerabilities in operational infrastructure. Other prominent cases include the 2019 disruption of communications with multiple renewable installations in Utah, and recent attacks on Poland’s electricity grid [9], in which adversaries bypassed local firewalls and erased supervisory device firmware without

forensic traces. Beyond disrupting communication and supervisory functions, vulnerable SCADA interfaces can also be exploited to target grid stability directly. This is exemplified in [10], [11], where the control parameters of flexible AC transmission system (FACTS) devices are manipulated to alter the reactance of multiple transmission lines, compromising the grid’s frequency stability.

In this context, PLLs emerge as particularly vulnerable targets, as they strongly influence IBR transient behavior through their interactions with multiple control loops [12], [13]. This vulnerability is further exacerbated by revised grid codes such as IEEE 1547 [14], which now mandate that GFL converters, previously focused on market participation and maximum power point tracking, implement primary control loops to support grid frequency and voltage, similar to grid-forming (GFM) IBRs [15]. Meeting these requirements relies on accurate frequency estimation, conventionally derived from PLLs [16]. As a result, PLLs are no longer used strictly for synchronization but also support controllers spanning longer time-scales, such as droop control, thereby dictating the overall small-signal profile of IBRs. This strong coupling between PLLs and the various control layers may be exploited by adversaries seeking to harm this critical infrastructure: as illustrated in [17], attackers exploiting vulnerabilities in IBR communication channels can gain access to supervisory interfaces and manipulate critical PLL gains to degrade system operation, destabilize the network, or unnecessarily trigger protection mechanisms.

The severity of such attacks lies in their stealthy nature. As shown in [17], if PLL tampering occurs at steady state, the system may appear unaffected, with the consequences manifesting only during subsequent disturbances. However, that work does not analyze the underlying mechanisms that render these attacks stealthy and evasive, and thus difficult to detect, which warrant further investigation. The same threat model is examined in [18], where adversaries nullify the integral gain of the synchronization PLL to induce power deviations, exploiting the ability of the resulting proportional-only PLL to track grid frequency while introducing a steady-state phase error that produces power regulation errors in current-controlled GFL IBRs. However, this attack is conspicuous rather than stealthy, as the loss of PLL phase lock leaves an immediate detection trace and is readily flagged by protection mechanisms. Furthermore, power shifting through this threat model can only be realized in the absence of dedicated primary controllers, a highly unlikely scenario given the mandate of IEEE 1547 [14]. To mitigate this vulnerability, the authors propose an alternative synchronization scheme that neutralizes

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the attack at the expense of degraded system performance. The authors further introduce an attack detection method; however, it is restricted to the case in which the integral gain is nullified and fails to detect any other PLL gain tampering pattern. Similarly, [19] presents a PLL-less current control strategy that addresses the considered threat model but does not surpass the performance of conventional PLL-based implementations. These limitations establish a clear need for a systematic framework capable of rigorously analyzing this attack vector, quantifying its impact on system performance, and exposing otherwise stealthy intrusions, while providing non-disruptive protection for vulnerable power electronic interfaces.

The stakes of this gap are high: without an explicit characterization of how PLL tampering propagates through the control layers, operators cannot gauge an attack's severity or the conditions under which its consequences surface, a pressing concern for grid-supporting GFL IBRs that rely on PLLs for primary-control implementation. In [17], [18], simulation-based analyses of grid-supporting IBRs highlight the impact of estimator dynamics on system performance and fault recovery time. However, an explicit small-signal analysis incorporating these supplementary loops is still lacking, particularly under weak-grid conditions where the control loops interact with the PLL used for synchronization. These analyses are essential to fully understand these interactions and to ensure that IBR infrastructures remain stable and resilient to poor transients and potential instabilities.

This paper pursues two primary objectives. First, it formulates and rigorously analyzes a mathematical threat model for stealthy PLL tampering, explicitly capturing the mechanism by which the attack evades detection and quantifying its impact on IBR dynamics through comprehensive small- and large-signal analyses. Prior works study such attacks, but do not analyze the mechanism underlying their stealthiness. Second, an effective detection framework is developed to safeguard critical IBR infrastructure without compromising nominal control performance, making it straightforward to integrate into existing GFL plants. Specifically, this work proposes a novel robust synchronization unit (RSU)-based PLL architecture that renders both proportional and integral gain manipulations observable through shifts in equilibrium points while preserving the small-signal behavior of conventional PLLs. This overcomes the limitations of existing approaches, which either detect only the special case of integral-gain nullification, degrade performance, or eliminate the PLL altogether, as summarized in the comparison in Table I. A further strength of the framework is its generality, as the same equilibrium-based principle can be extended naturally to a broad class of IBR control architectures, exposing parameter changes wherever they occur. Its effectiveness is demonstrated through extensive simulation studies and validated via controller hardware-in-the-loop (CHIL) experiments.

The remainder of this article is organized as follows. Section II presents a detailed modeling framework for IBR-dominated power systems, while Section III introduces a grid-code-compliant GFL control framework based on dynamic frequency estimation. The threat model is formulated in Section IV, and Section V provides both small- and large-signal

TABLE I: Comparison with the existing literature.

| Feature                                                       | [17] | [18] | [19] | This work |
|---------------------------------------------------------------|------|------|------|-----------|
| Threat model and stealthiness analysis                        | ○    | ○    | ○    | ✓         |
| Small-/large-signal attack impact assessment                  | ○    | ×    | ○    | ✓         |
| PLL gain tampering detection                                  | ×    | ○    | ○    | ✓         |
| Nominal PLL performance preserved                             | ✓    | ×    | ×    | ✓         |
| Industrial/GFL controller compatibility                       | ✓    | ×    | ×    | ✓         |
| ✓ fully addressed    ○ partially addressed    × not addressed |      |      |      |           |

analyses to examine the interactions among the frequency estimator, synchronization and control layers, and other grid-connected resources. The proposed detection scheme is presented in Section VI, and experimental results validating its effectiveness and applicability are reported in Section VII. Finally, Section VIII concludes the paper.

## II. SYSTEM MODELING

A model-based approach is required to comprehensively analyze PLL-targeting cyber-attacks, as adversarial gain tampering ultimately seeks to alter the small-signal characteristics of IBRs. Such analyses enable the approximate quantification of the impact of these cyber-attacks and facilitate the formulation of the most detrimental attack models, which inherently supports the design of effective detection schemes. Accordingly, this section presents a detailed model of IBR-dominated networks, comprising a diverse mixture of GFL and GFM resources and their associated control schemes.

### A. Power Converters

Power converters are commonly represented using equivalent averaged models, in which they are formulated as ideal voltage sources connected to the point of common coupling (PCC) through an RL filter and a shunt capacitor. This system can be expressed using the Park transformation, which maps system states into a stationary two-axis ( $dq$ ) reference frame, thereby facilitating analysis and control design. The current dynamics in the  $dq$  domain are given by [20]:

$$l_i \dot{v}_{d,i}^{L_i} = -r_i v_{d,i}^{L_i} + \omega_{p,i} l_i v_{q,i}^{L_i} + v_{f,d,i}^{L_i} - v_{c,d,i}^{L_i}, \quad (1)$$

$$l_i \dot{v}_{q,i}^{L_i} = -r_i v_{q,i}^{L_i} - \omega_{p,i} l_i v_{d,i}^{L_i} + v_{f,q,i}^{L_i} - v_{c,q,i}^{L_i}, \quad (2)$$

where  $i \in \mathcal{C}$ , with  $\mathcal{C}$  being the set of power converters. The parameters  $r_i$ ,  $l_i$  denote the  $i$ th converter parasitic resistance and inductance of the filter,  $\omega_{p,i}$  is the IBR frequency, derived from PLLs for GFL converters and from the swing equation for GFM ones,  $v_{d,i}^{L_i}$ ,  $v_{q,i}^{L_i}$  are the converter output currents,  $v_{f,d,i}^{L_i}$ ,  $v_{f,q,i}^{L_i}$  and  $v_{c,d,i}^{L_i}$ ,  $v_{c,q,i}^{L_i}$  are the  $dq$ -frame inverter and capacitor voltages. The superscript  $(\cdot)^{L_i}$  denotes quantities expressed in the  $i$ th local reference frame. The capacitor voltages of the power converters' filters are given by:

$$C_i \dot{v}_{c,d,i}^{L_i} = -v_{c,d,i}^{L_i}/R_i + \omega_{p,i} C_i v_{c,q,i}^{L_i} + \sum_{i \in \Omega_j} v_{in,d,i}^{L_i} - \sum_{i \in F_j} v_{out,d,i}^{L_i}, \quad (3)$$

$$C_i \dot{v}_{c,q,i}^{L_i} = -v_{c,q,i}^{L_i}/R_i - \omega_{p,i} C_i v_{c,d,i}^{L_i} + \sum_{i \in \Omega_j} v_{in,q,i}^{L_i} - \sum_{i \in F_j} v_{out,q,i}^{L_i}, \quad (4)$$

with  $C_i$ ,  $R_i$  denoting the capacitance and resistive load, and  $v_{c,d,i}^{L_i}$ ,  $v_{c,q,i}^{L_i}$  the capacitor  $dq$ -voltages. The summed terms correspond to the line currents entering and leaving the capacitor. Specifically,  $\Omega_j$  and  $F_j$  denote, respectively, the sets of indices associated with currents entering and leaving the capacitor.

The voltage and current equations are defined in the local  $dq$  frame associated with each power converter. Specifically, each subsystem operates in its own  $dq$  frame, which enables the implementation of its controller and requires all system states to be expressed in the corresponding local coordinates.

### B. Line, Load and Network Dynamics

The power network is modeled using strictly resistive-inductive lines and complex loads. Contrary to [17], this paper considers the full nonlinear models of both lines and loads in order to accurately capture their interactions and their impact on the stability of IBRs. The dynamics of the transmission lines are described by:

$$l_{l,i} i'_{l,d,i}^G = -r_{l,i} i_{l,d,i}^G + \omega_g l_{l,i} i_{l,q,i}^G + v_{s,d,i}^G - v_{e,d,i}^G, \quad (5)$$

$$l_{l,i} i'_{l,q,i}^G = -r_{l,i} i_{l,q,i}^G - \omega_g l_{l,i} i_{l,d,i}^G + v_{s,q,i}^G - v_{e,q,i}^G, \quad (6)$$

where  $i \in \mathcal{L}$ , with  $\mathcal{L}$  denoting the set of transmission lines. Here,  $i'_{l,d,i}$  and  $i'_{l,q,i}$  denote the line currents in the  $dq$  reference frame;  $r_{l,i}$  and  $l_{l,i}$  are the line resistance and inductance, respectively; and  $v_{s,d,i}^G$ ,  $v_{s,q,i}^G$  and  $v_{e,d,i}^G$ ,  $v_{e,q,i}^G$  represent the sending- and receiving-end voltages in the  $dq$  frame. The superscript  $(\cdot)^G$  denotes quantities expressed in the global reference frame. The global reference frame is a common rotating frame with angular frequency  $\omega_g$  and is used to mathematically interconnect the local reference frames of the converters. At steady state, it represents the grid frequency. Inductive loads can be modeled in the same manner by assuming that the receiving-end voltage of the corresponding line is connected to ground. Converter-free load buses can also be modeled using (3)-(4); however, in this case, all states must be formulated in the global reference frame.

### C. Generalized Power System Model

The distinction between local and global reference frames enables the modeling of synchronization dynamics for individual subsystems, including GFL and GFM IBRs. However, it complicates stability analysis, as transitioning between reference frames requires coordinate transformations. Suppose that  $x^{L_i} = [x_d^{L_i}, x_q^{L_i}]$  and  $x^G = [x_d^G, x_q^G]$  represent a power-system state expressed in the local and global reference frames, respectively. These states can be written in phasor form as:

$$X^G = (x_d^G + jx_q^G) e^{j\int \omega_g dt}, \quad (7)$$

$$X^{L_i} = (x_d^{L_i} + jx_q^{L_i}) e^{j\int \omega_{p,i} dt}. \quad (8)$$

This representation is consistent with the  $\alpha\beta$  transformation; therefore, a rotation matrix,  $R$  [21], can be used to transition between reference frames. The same concept applies to the phasor representation, yielding:

$$X^G = X^{L_i} e^{j\int (\omega_{p,i} - \omega_g) dt}, \quad (9)$$

or, equivalently:

$$X^G = (x_d^{L_i} + jx_q^{L_i}) e^{-j\Delta\theta}, \quad (10)$$

where  $\Delta\theta$  denotes the phase difference between the local and global frames and satisfies:

$$\Delta\dot{\theta}_i = \omega_g - \omega_{p,i}. \quad (11)$$

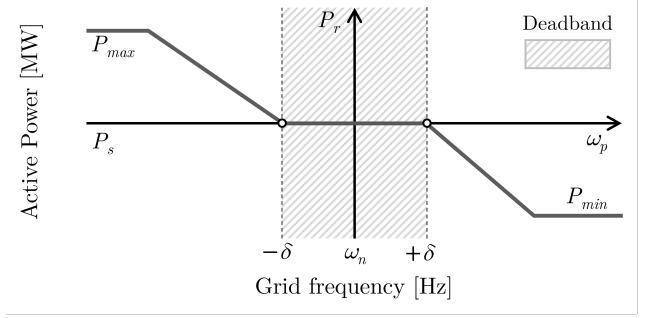


Fig. 1: Frequency-support curve.

Similarly, the transformation from the global reference frame to the local reference frame is given by:

$$X^{L_i} = (x_d^G + jx_q^G) e^{j\Delta\theta}. \quad (12)$$

At steady state,  $\omega_g = \omega_{p,i}^s$  must hold for all  $i \in \mathcal{C}$ , where the superscript  $(\cdot)^s$  denotes steady-state, to ensure that the  $dq$  quantities expressed in the global reference frame remain constant. This requirement facilitates stability analysis, as the corresponding equilibrium point is a constant vector, provided that the power system model is stable. It is emphasized that even if the reference frequency of the global frame differs from that of the local frames at steady state, the local  $dq$  frames constructed by the power converters will still reach consensus on a common system frequency as a result of primary control. In this study, the global reference frequency  $\omega_g$  is derived from the local reference frame of one of the power converters.

### III. ANCILLARY SERVICES-ORIENTED IBR CONTROLLERS

Naturally, GFM converters provide ancillary services such as frequency and voltage support. However, GFL IBRs mainly focus on maximum power injection. On the other hand, conventional GFL schemes are not designed explicitly for real-time frequency support and can occasionally weaken the grid instead of strengthening it. To address this concern, revised grid regulations, [14], require GFL converters to strictly follow the droop-like characteristic illustrated in Fig. 1, where the power output of the converter is adjusted according to the system's frequency. These curves exhibit two key features: the *deadband* and the *slope*, the latter being equivalent to the active-power droop coefficient. Similarly, an analogous droop characteristic is specified for the reactive power control loop with respect to the PCC voltage; however, it is omitted here to avoid redundancy. For clarity, the converter index  $i$  is suppressed hereafter, as the following analysis focuses on a single representative GFL converter, with the results applying identically to every  $i \in \mathcal{C}$ . The mathematical representation of the grid-support controller is given below:

$$P_r = P_s + f(m_P(\omega_n - \omega_g)), \quad (13)$$

$$Q_r = Q_s + f(m_Q(|v|_n - |v|_{pcc})), \quad (14)$$

where  $\omega_g$  and  $\omega_n$  correspond to the measured and nominal grid frequencies,  $v_n$  and  $v_{pcc}$  the nominal and measured PCC voltage. Finally,  $m_P$ ,  $m_Q$  correspond to the active and reactive power droop gains, while  $f$  denotes the deadband function:

$$f(x) = \begin{cases} x - \delta, & \text{if } x > \delta, \\ 0, & \text{if } |x| \leq \delta, \\ x + \delta, & \text{if } x < -\delta, \end{cases} \quad (15)$$

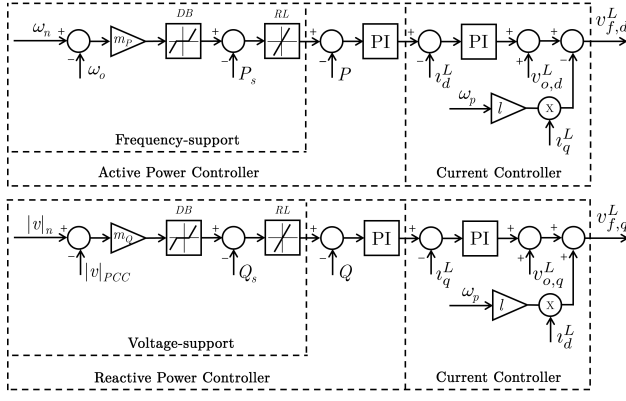


Fig. 2: Grid-supporting GFL control scheme.

where  $\delta$  represents the deadband threshold. According to the ENTSO-E Frequency Sensitive Mode grid code [22], the maximum value for  $\delta$  is 0.5 Hz. For  $\delta = 0$ , the deadband is always inactive.

The reference power is then usually channeled through a rate limiter (RL) to ensure that it adheres to physical plant restrictions and grid-codes:

$$\dot{P}_{sr} = \dot{P}_m \text{sat}(a(P_r - P_{sr})), \quad (16)$$

where 'sat' denotes the saturation function,  $\dot{P}_m$  the maximum allowed derivative limit and  $a$  is a tuning parameter. The overall control diagram is illustrated in Fig. 2.

The controller defined by (13) and (16) requires two inputs: the reference power  $P_s$  and an estimate of the grid frequency. The method used to obtain  $\omega_g$  significantly affects the converter's response during frequency support. GFL converters implement PLLs for both synchronization and frequency estimation for grid-monitoring purposes. However, PLLs are tuned to act instantaneously, enabling the implementation of upper control layers. Consequently, the slightest grid-side disturbance significantly excites the PLL dynamics and its output frequency, which can drive an excessive reaction of the ancillary-service power controller despite the implemented deadband safeguard. This can be addressed by separating the synchronization and frequency-estimation functions into two distinct PLLs that are tuned individually, so that a slower frequency-estimation PLL can be assigned independently of the synchronization dynamics. The same flexibility applies in the opposite case: under weak-grid conditions, where the synchronization PLL must be slowed to preserve stability, the separated structure still permits a responsive frequency-estimation PLL, decoupling the frequency-support response from the synchronization bandwidth.

This representation is made without loss of generality, since it is structurally equivalent to a single synchronization PLL augmented with additional frequency-estimation loops. Expressing the scheme as two PLLs additionally allows each function, synchronization and frequency estimation, to be described and tuned in common PLL terms, such as bandwidth and damping ratio, so that the two loops are decoupled and the transient behavior and stability margins of each are inferred directly from well-established PLL design principles, as substantiated by the formal stability analysis in Section V.

Specifically, the synchronization layer implements the

synchronously rotating frame (SRF) PLL to track the phase, frequency and voltage magnitude of the grid:

$$\omega_p = \omega_n + k_{p,s} v_{c,q}^L + z, \quad (17)$$

$$\dot{z} = k_{i,s} v_{c,q}^L, \quad (18)$$

with  $k_{p,s}$  and  $k_{i,s}$  being the proportional and integral gains of the PLL, and  $\theta_p = \int \omega_p dt$  is the estimated input voltage phase. Since the analysis applies identically to both PLLs, the subscript  $s$  is omitted hereafter. The local-frame representation of the  $q$ -axis converter output voltage,  $v_{c,q}^L$ , can then be formulated using the transformation (12):

$$v_{c,q}^L + jv_{c,d}^L = (v_{c,d}^G + jv_{c,q}^G) e^{j\Delta\theta}, \quad (19)$$

where the auxiliary state  $\Delta\theta$ , representing the phase error between the global and local frames, is given by:

$$\Delta\dot{\theta} = \omega_g - \omega_p. \quad (20)$$

Equating the real and imaginary parts yields:

$$v_{c,q}^L = v_{c,q}^G \cos \Delta\theta + v_{c,d}^G \sin \Delta\theta. \quad (21)$$

The outer-loop controller employs the same PLL formulation to estimate the grid frequency:

$$\omega_o = \omega_n + k_{p,o} v_{c,q}^{Lo} + z_o, \quad (22)$$

$$\dot{z}_o = k_{i,o} v_{c,q}^{Lo}, \quad (23)$$

where  $k_{p,o}$  and  $k_{i,o}$  denote the proportional and integral gains of the frequency estimator, respectively. Here,  $\omega_o$  is the estimated frequency. The superscript  $(\cdot)^{Lo}$  indicates quantities expressed in the reference frame of the outer-loop controller.

Following the same procedure used for the synchronization PLL, the outer-loop representation of the PCC voltage is obtained by transforming it from the local PLL reference frame to the outer-loop reference frame:

$$v_{c,d}^{Lo} + jv_{c,q}^{Lo} = (v_{c,d}^L + jv_{c,q}^L) e^{j\Delta\theta_o}. \quad (24)$$

Here,  $\Delta\theta_o$ , similarly to  $\Delta\theta$ , is an auxiliary variable that represents the phase difference between the local reference frame and the outer-loop reference frame and is defined by:

$$\Delta\dot{\theta}_o = \omega_p - \omega_o. \quad (25)$$

Incorporating the ancillary services controller fundamentally alters the core operating principles of GFL converters. In this context, a critical role is played by the frequency estimator, and its interaction with the synchronization PLL, as well as with other grid components such as GFM converters, RLs and the deadband threshold. These interdependencies may be exploited by adversaries seeking to degrade or potentially destabilize power networks by manipulating PLL gains that are accessible through supervisory control interfaces. To fully assess the potential impact of such malicious actions, a comprehensive threat model must be formulated to characterize both the nature and the consequences of the cyber-attack.

#### IV. ATTACK MODEL FORMULATION

Building on the vulnerability exposed in the previous section, this section formalizes the attacker's objective and the notion of stealthiness underlying PLL gain-tampering attacks. Even when the frequency-estimation and synchronization PLLs are properly tuned offline, both remain exposed to deliberate manipulation by adversaries who compromise

supervisory control interfaces [6], overwriting the proportional and integral gains of either loop. These interfaces, typically SCADAs and PLCs, can reportedly be breached by exploiting software vulnerabilities to obtain write access, whether through low-privilege accounts or remotely without prior system access, and to overwrite critical control parameters [7], [23]. The severity of this attack lies in its stealthy nature: altering the gains once the system has reached steady state produces no observable anomaly in the nominal response of the PLLs, and the impact surfaces only when a subsequent disturbance excites the modified dynamics. The attack is practically manifested as a one-time parameter overwrite that requires no real-time observation by the adversary, a threat model also considered in [17]–[19], and examined next to identify the factors that allow it to remain undetected.

Before proceeding with the analysis, the PLL gains prior to the attack are denoted as  $k_p^n$  and  $k_i^n$ , and the gains after the attack as  $k_p^a$  and  $k_i^a$ . The same analysis applies to both the frequency estimator and the synchronization PLL; hence, the gain subscripts are omitted, and the analysis is conducted for a generic PLL model. Accordingly, the objective of the attacker can be summarized as follows:

$$k_p^n \rightarrow k_p^a, \quad k_i^n \rightarrow k_i^a. \quad (26)$$

To analyze how an attack is perceived from the operator's perspective, the closed-loop PLL dynamics are reformulated by incorporating the coordinate transformation in (20) and assuming interconnection to an arbitrary grid voltage. Suppose that the operator has access to  $v_{c,q}^{Lo}$  and the frequency estimate provided by the PLL. For  $\hat{V} = 1$  pu, the  $q$ -axis voltage in the local frame is given by  $v_{c,q}^L = \hat{V} \sin \Delta\theta$ , yielding:

$$\Delta\dot{\theta} = \omega_g - \omega_n - k_p^n \sin \Delta\theta - z, \quad (27)$$

$$\dot{z} = k_i^n \sin \Delta\theta. \quad (28)$$

The equilibrium points are yielded from:

$$\Delta\theta^s = 0, \quad z^s = \omega_g - \omega_n. \quad (29)$$

An attack is *stealthy* if operators or monitoring systems that check the states against their expected steady-state values cannot detect it. Whether the adopted threat model is stealthy thus depends on its effect on the equilibrium. Let  $\eta^s = (\Delta\theta^s, z^s)^n - (\Delta\theta^s, z^s)^a$  be the difference between the pre- and post-attack equilibria, where  $(\Delta\theta^s, z^s)^n$  and  $(\Delta\theta^s, z^s)^a$  denote the equilibrium pairs before and after the attack, as monitored by the operator. From (29), tampering that preserves integral action leaves the equilibrium unchanged, so  $\eta^s = 0$ , in agreement with the observations in [17], [18]. Consequently, if the system is at steady state when (26) occurs, it remains at equilibrium, unperturbed and without any transient behavior, despite the changed gains.

A straightforward and conspicuous strategy that an adversary may adopt is to completely nullify the integral gain ( $k_i^a = 0$ ), as illustrated in [18]. As shown in [24], the SRF PLL can still track the grid frequency with a proportional-only controller, that is, with the integral gain set to zero; however, exact phase tracking cannot be achieved, resulting in a steady-state error in power regulation when direct power control is implemented through current regulators, as in [19]. Under this

condition, an attacker can actively manipulate the active power output of the plant [19]. Here, since  $\eta^s = [(\omega_g - \omega_n)/k_p^a, 0] \neq [0, 0]$ , the attack is immediately detected: the induced phase error prevents the exact phase lock that synchronization requires, which is reflected in critical controller states such as the  $dq$  voltages and triggers converter disconnection due to loss of synchronization, exposing the tampering. The more capable threat model of [17], however, which considers all possible combinations of PLL gain manipulation that preserve integral action, always yields  $\eta^s = 0$ , leaving no observable trace for operators to detect the attack, which makes it particularly dangerous and is therefore the threat model considered in this work. Since  $\eta^s$  captures precisely any shift in the monitored steady-state values, it provides an exact mathematical characterization of the stealth notion introduced above.

**Remark 1.** A PLL gain tampering cyber-attack remains stealthy if and only if  $\eta^s = 0$ .

Consequently, capturing any tampering of the PLL gains requires  $\eta^s \neq 0$ , a condition that, for the threat model of (26), cannot be satisfied by the conventional PLL structure. This becomes clearer through (29): owing to the exact phase tracking enabled by the integral term, the equilibrium point is always the same for a given grid frequency. This motivates a systematic analysis of the impact of both the frequency-estimation and synchronization PLL parameters on the small- and large-signal behavior of the IBR, in order to anticipate gain-tampering strategies that degrade system performance and to design a detection mechanism capable of identifying such attacks.

## V. PLL IMPACT ON STABILITY AND TRANSIENT RESPONSE

As evident from (13), the frequency estimator and its interaction with nonlinear elements, such as RLs, critically influence the dynamic response and small-signal behavior of GFL converters. To capture these effects, this section conducts an explicit small-signal stability analysis, complemented by a comprehensive simulation study that examines the relationship between the quality of grid-supporting services provided by the IBR and the frequency estimator.

### A. Small-Signal Analysis

To analyze the interaction between the frequency-supporting GFL converter and the grid, a simplified single-bus model is considered, in which a GFM converter is connected to both

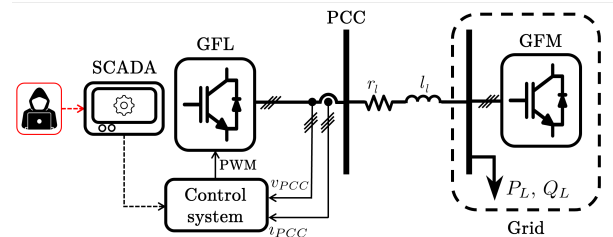
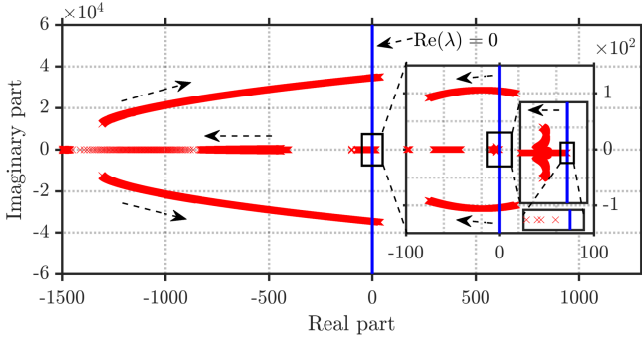
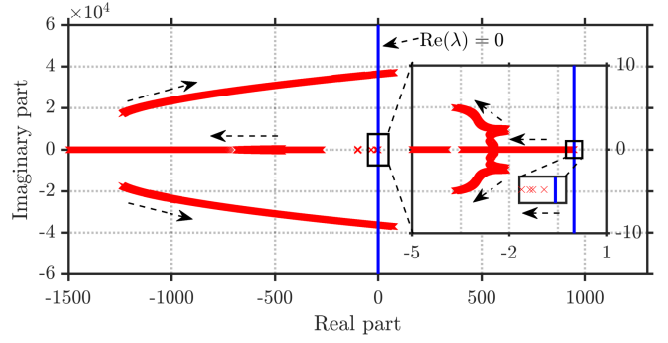


Fig. 3: GFL converter connected to a dynamic grid.



(a) Synchronization PLL bandwidth: 40 Hz



(b) Synchronization PLL bandwidth: 1 Hz

Fig. 4: Modal analysis for a GFL converter connected to a GFM converter through a 2.5 mH line, for different PLL bandwidths.

a GFL converter and a load at the same bus. The simplified swing equation can be expressed as:

$$J\Delta\dot{\omega} = -D\Delta\omega + P_{GFM} + P_{GFL} - P_L, \quad (30)$$

where  $\Delta\omega = \omega - \omega_n$  is the frequency deviation of the GFM converter, and  $P_{GFM}$ ,  $P_{GFL}$ , and  $P_L$  denote the power outputs of the GFM, GFL converters, and the load, respectively. The parameter  $D$  is the droop coefficient of the GFM IBR and  $J$  its virtual inertia.

Although modern grid codes require GFL converters to follow the reference structure in (16), here (13) is used for analytical clarity. Assuming that the PLL input voltage is 1 pu, the  $q$ -axis voltage used by the PLL outer loop in the local reference frame becomes:

$$v_q^{Lo} = \sin\Delta\phi. \quad (31)$$

where  $\Delta\phi$  is given by:

$$\Delta\dot{\phi} = \omega - \omega_o, \quad (32)$$

the PLL dynamics can be written as:

$$\Delta\dot{\phi} = \Delta\omega - k_{p,o}\sin\Delta\phi - z_o, \quad (33)$$

$$\dot{z}_o = k_{i,o}\sin\Delta\phi. \quad (34)$$

where the superscript  $(\cdot)^n$ , denoting pre-attack quantities, is omitted for clarity.

Substituting these dynamics into the swing equation and linearizing the trigonometric terms around the equilibrium  $\Delta\phi^* = 0$ , the closed-loop linearized system is:

$$J\Delta\ddot{\omega} = -D\Delta\tilde{\omega} + P_{GFM} + P_s - P_L - m_P(k_{p,o}\Delta\tilde{\phi} + k_{i,o}\tilde{z}_o), \quad (35)$$

$$\Delta\dot{\tilde{\phi}} = \Delta\tilde{\omega} - k_{p,o}\Delta\tilde{\phi} - k_{i,o}\tilde{z}_o, \quad (36)$$

$$\dot{\tilde{z}}_o = \Delta\tilde{\phi}. \quad (37)$$

The state matrix of this system is:

$$\begin{bmatrix} -D/J & -m_P k_{p,o}/J & -m_P k_{i,o}/J \\ 1 & -k_{p,o} & -k_{i,o} \\ 0 & 1 & 0 \end{bmatrix}.$$

Applying the Routh-Hurwitz criterion to obtain the condition for the eigenvalues to have strictly negative real parts yields the stability condition:

$$\frac{m_P k_{i,o}}{J} - k_{p,o} k_{i,o} < \frac{k_{p,o}^2 D}{J} + \frac{m_P k_{p,o}^2}{J} + k_{p,o} \left( \frac{D}{J} \right) + \frac{D m_P k_{p,o}}{J^2}. \quad (38)$$

This inequality is not satisfied by every combination of PLL or network parameter values, meaning the system can become unstable for poorly chosen parameters. Notably, when  $k_{p,o} =$

0, a conventional PLL would be marginally stable, yet under this simplified model the overall system becomes unstable. If instead the GFL converter is connected to an infinite bus,  $J \approx \infty$ , the stability condition reduces to  $-k_{p,o} k_{i,o} < 0$ , which is always satisfied. In this case, the one-bus model explicitly reveals that the two subsystems decouple, guaranteeing system stability independently of their interaction. To obtain a more interpretable stability condition, (38) is rewritten as:

$$\frac{m_P k_{i,o}}{J} < k_{p,o} \left( k_{i,o} + \frac{k_{p,o} D}{J} + \frac{m_P k_{p,o}}{J} + \left( \frac{D}{J} \right)^2 + \frac{D m_P}{J^2} \right). \quad (39)$$

This inequality indicates that when the PLL is extremely under-damped, the combined system may readily become unstable. The simplified GFM model is stable in stand-alone operation; therefore, any potential instability arises solely from the GFL converter and its outer-loop PLL. A similar observation was reported in [25], where interactions between GFM and GFL converters at low bandwidths were shown to induce instability. This instability is strongly influenced by the frequency estimator dynamics and may be exploited by adversaries to degrade system performance or, in extreme cases, destabilize the power system.

Although the previous results indicate strong coupling among the IBRs, the reduced-order model cannot fully capture their interactions, owing to simplifying assumptions such as neglecting essential IBR dynamics, including the current and power control loops of the primary (voltage-frequency) control shown in Fig. 2, and, most importantly, the impact of the synchronization PLL. To address this limitation, an eigenvalue analysis is performed on the testbench shown in Fig. 3 under both strong and weak grid conditions, using the complete nonlinear closed-loop model established in Sections II and III, incorporating the network dynamics and the current and power control loops. The eigenvalues are obtained directly from the Jacobian of this model, evaluated at the system's equilibrium point [13]. The GFL converter is rated at 0.1 MVA, whereas the GFM IBR has a rating of 1 MVA, with all droop coefficients for both converters set to 1%.

First, a weak-grid is considered with a 2.9 SCR index. The synchronization PLL bandwidth is set to 40 Hz, and the eigenvalues of the system are computed as the bandwidth of the frequency estimator is increased from 0 to 10 Hz. The damping ratio  $\zeta$ , is fixed at  $1/\sqrt{2}$ . The results are shown in Fig. 4a, where the arrows indicate the migration of the

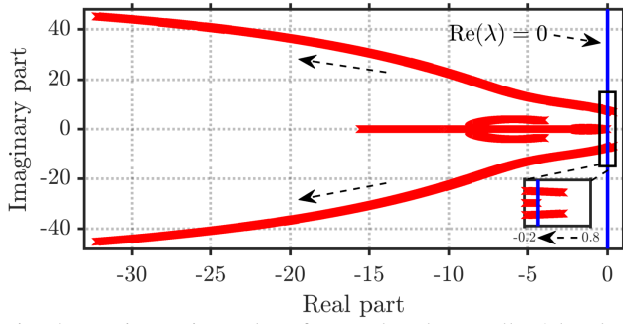


Fig. 5: Dominant eigenvalues for a reduced controllers' bandwidth.

eigenvalues as the PLL bandwidth increases. It can be observed that the system is unstable for lower bandwidth values, primarily due to weak-grid conditions associated with the low SCR. As the estimator bandwidth increases, the frequency-support controller stabilizes the system. This stabilizing effect is attributed to the grid-forming characteristics of the controller and its contribution to regulating the PCC frequency and voltage. However, further increasing the bandwidth eventually destabilizes the system, as the power and current controllers as well as the PLL are unable to follow in time the output of the ancillary services controller. To further illustrate this phenomenon, the controller bandwidths are reduced by a factor of three, and the same analysis is repeated. The results, shown in Fig. 5, indicate that without the frequency-support mechanism the system remains unstable; however, as the frequency estimator bandwidth increases, stability is achieved at a lower frequency compared to Fig. 4a. Moreover, the system remains stable over a wider range of estimator bandwidths. Overall, excessively slow tuning of the frequency estimator limits the frequency-support capability of the IBRs and can lead to instability under weak-grid conditions, while overly aggressive tuning may produce similar destabilizing effects.

Next, the synchronization PLL bandwidth is reduced to 1 Hz, and the experiment is repeated. The results in Fig. 4b demonstrate that this adjustment mitigates both small-signal instability and low-frequency oscillatory modes. In contrast to using the synchronization PLL for frequency estimation, increasing the bandwidth of a dedicated PLL now enhances the fault-ride-through capability of the IBR, as the dominant eigenvalues shift further into the left-half plane. However, further increasing the PLL bandwidth again leads to instability. Finally, the transmission line is removed and the GFM converter is replaced by an infinite bus. The analysis is repeated for both PLL bandwidth configurations, and the dominant eigenvalues are shown in Fig. 6. As expected, the dominant eigenvalues coincide in both cases, and the system remains stable for all PLL bandwidths.

These results underscore the critical role of frequency support and estimator dynamics in system stability. Improper gain tuning can degrade performance or even induce instability, providing a potential avenue for adversaries to stealthily compromise the small-signal behavior of IBRs and the broader power system.

### B. Large-Signal Analysis

This section investigates the impact of the frequency estimator dynamics on the performance of the grid-supporting

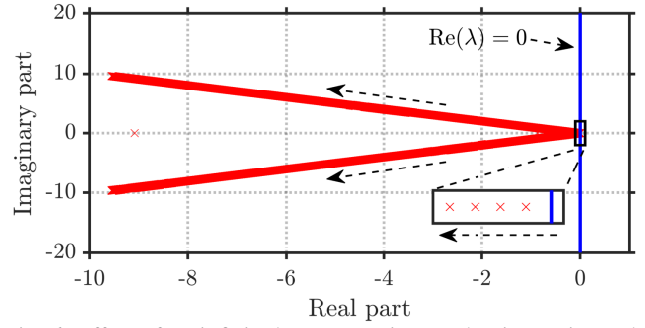


Fig. 6: Effect of an infinite bus connection on dominant eigenvalues.

IBR during large-signal disturbances, such as load changes. To conduct a more holistic analysis, a grid with four IBRs is considered, as shown in Fig. 7.

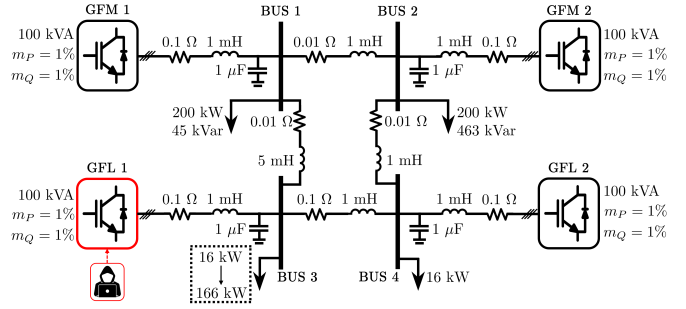
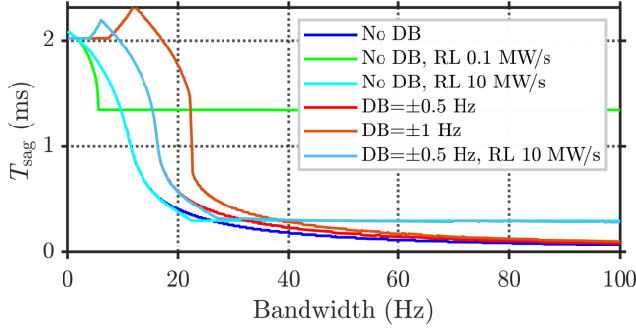


Fig. 7: Simulation benchmark.

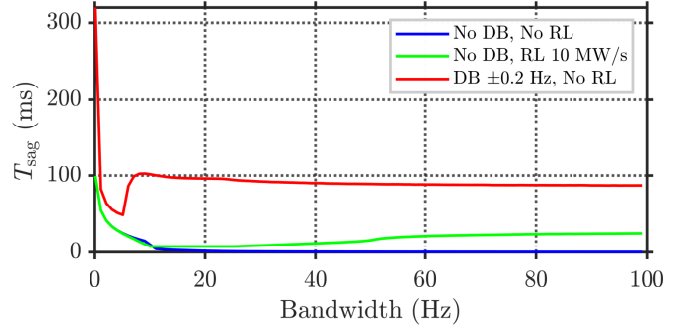
The performance of the frequency-support mechanism is evaluated by measuring the duration of a voltage sag induced by a significant load change at bus 3 (+0.15 MW), where the under-study grid-supporting IBR is located. Specifically, the voltage-sag duration is measured for different deadband settings, RL derivative limits, and frequency-estimator bandwidths. The results are illustrated in Fig. 8a.

Each primary controller configuration yields a distinct dynamic response, which may either improve or degrade the voltage recovery time. The shortest recovery times are obtained when no deadband is applied, allowing the controller to respond immediately to the disturbance. Introducing a RL with tight power-derivative constraints significantly improves the response for low PLL bandwidths by suppressing excessively aggressive control actions. For higher bandwidths, the recovery time remains nearly constant, as the high-derivative response of the primary controller is constrained by the RL. Once the RL becomes inactive, the frequency estimator has already converged to the grid frequency and therefore has no further impact on the short-term system dynamics. Increasing the derivative limits results in similar behavior; however, the saturation of the recovery time occurs at a substantially shorter duration.

Incorporating a deadband in the controller limits preemptive control actions and reduces the participation of the IBR in grid-support services. As observed in Fig. 8a, the converter does not contribute to voltage recovery when the estimator bandwidth is below 4 Hz and the deadband is set to  $\pm 0.5$  Hz. For bandwidths between 4 and 9 Hz, the controller operates counter-productively and increases the recovery time, whereas for higher bandwidths it improves the response. Increasing



(a) Inductive network.



(b) Resistive network.

Fig. 8: Voltage sag duration for different controller and PLL parameters.

the deadband to  $\pm 1$  Hz leads to qualitatively similar but more pronounced effects. The combined use of RLs and deadbands produces an interaction in which the controller degrades performance at low estimator bandwidths, improves recovery at intermediate bandwidths, and ultimately leads to saturation of the recovery time at high bandwidths due to the restrictive action of the RL.

The impact of the frequency estimator is more pronounced in resistive networks. To further examine this effect, the experiment is repeated for a predominantly resistive system ( $X/R = 0.6$ ), where both lines and loads exhibit mainly resistive characteristics. As shown in Fig. 8b, the frequency estimator plays a critical role in disturbance rejection, with recovery times ranging from 0.1 ms to 300 ms. However, its interaction with nonlinear protection elements may either improve or degrade recovery performance. This behavior is most evident when the grid-support controller employs RLs: lower bandwidths reduce recovery time, whereas, in contrast to inductive networks, increasing the bandwidth results in a gradual increase in recovery time. These results highlight the influence of the frequency estimator and its dynamic response on fast frequency support services. While the combination of conventional primary control with nonlinear and discontinuous elements can improve system performance during disturbances, improper tuning may instead degrade the response. More importantly, deliberate or *malicious* manipulation of controller parameters can render the frequency-support mechanism counterproductive and may lead to instability of the IBR as well as adverse interactions with surrounding loads and resources.

## VI. RSU-BASED DETECTION OF PLL GAIN TAMPERING

A key conclusion derived from the threat-model impact assessment is that, to meaningfully affect the small-signal behavior of the plant, an adversary must significantly alter the proportional gain, the integral gain, or both, since properly tuned PLLs yield a system that is neither marginally stable nor underdamped. Building on this insight, this section presents an effective detection mechanism based on the RSU framework introduced in [24], capable of capturing stealthy PLL gain manipulations without compromising system performance. The proposed detection scheme relies on observing the internal states of an alternative PLL implementation that behaves identically to the conventional SRF PLL. The proposed PLL is described by:

$$\dot{\theta}_R = \omega_R = \omega_N + k_i z, \quad (40)$$

$$\dot{z} = \arctan v_{c,q}^R - k_p z, \quad (41)$$

with the Park transformation phase angle for the local control frame given by

$$\theta_P = \theta_R + k_p z. \quad (42)$$

Here  $\omega_N$  is a tunable constant, and the superscript  $(\cdot)^R$  denotes quantities expressed in the RSU internal reference frame. Additionally,  $\omega_R$  and  $\theta_R$  are RSU related internal states. An explicit illustration of the proposed module is presented in Fig. 9. Considering the coordinate transformation in (20), the pre-attack equilibrium point of the RSU is:

$$\Delta\theta^s = k_p^n (\omega_g - \omega_N) / k_i^n, \quad z^s = (\omega_g - \omega_N) / k_i^n, \quad (43)$$

whereas now  $\Delta\dot{\theta} = \omega_g - \omega_R$ . Note, at steady state, the phase error between the Park transformation phase and the grid voltage,  $\int (\dot{\theta}_P - \omega_g) dt$ , is zero. In contrast, due to the RSU structure, the internal phase error,  $\arctan v_{c,q}^R = \Delta\theta$ , is nonzero. Note that  $\omega_N$  is a constant feed-forward term, traditionally set to  $\omega_n$ , and therefore does not affect the small-signal dynamic response of the PLL. Conventionally, it reduces the tracking effort of the PLL by providing a frequency reference close to that of the grid; here, however,  $\omega_N$  assumes two additional roles that reinforce the proposed detection framework against the adopted threat model, as detailed below.

Evidently, the vector  $\eta^s$  now captures any PLL gain tampering, since the equilibrium points of (40) and (41) depend on both  $k_p^n$  and  $k_i^n$ , leaving no exploitable pattern for an adversary to mask an attack. Furthermore, any change in the controller gains induces a transient in the IBR response, so that even under communication tampering an anomalous, unexplained event is flagged. Beyond this,  $\omega_N$  serves two roles. First, it renders the communicated signals privacy-preserving: with  $\omega_N \neq \omega_n$ , (43) provides only two equations in the three unknowns  $k_p^n$ ,  $k_i^n$ , and  $\omega_N$ , so the gains cannot be fully reconstructed by an interceptor without knowledge of  $\omega_N$ ; the plant operator, which alone knows the implementation structure and  $\omega_N$ , remains the only entity able to recover them. Second, it sets the detection margin: since  $\omega_N$  scales the equilibrium points in (43), increasing  $|\omega_g - \omega_N|$  enlarges the steady-state offsets  $|\Delta\theta^s|$  and  $|z^s|$ , and hence the equilibrium shift produced by a gain change, making gain tampering easier to distinguish from frequency fluctuations, measurement

noise, and grid-side disturbances. Since  $\omega_N$  does not affect the transient response of the PLL, it can be tuned to achieve the desired detection margin and is set once at startup. Moreover, it need not be exposed to the supervisory (SCADA) interface, which keeps it private and beyond the reach of the adversary, who could otherwise mask only integral-gain tampering; since  $\Delta\theta^s = k_p^n z^s$  from (43) fixes the proportional relation between the two equilibria independently of  $\omega_N$ , any change in  $k_p^n$  alters this relation and exposes the attack. Alternatively,  $\omega_N$  can be periodically re-keyed through a cryptographic scheme, so that a leaked value can be exploited, for gain inference via (43) or attack masking, only until its next update.

The proposed scheme thus produces an equilibrium-based footprint that exposes any change in the PLL gains, prompting and enabling operators to investigate whether the change reflects malicious tampering or a legitimate manual or adaptive reconfiguration. This distinction is ultimately made by the plant operator; the proposed device reveals what was previously invisible, owing to the stealthy nature of the attack established in Section IV. Detection is governed by two parameters: an observation window  $T_w$ , which sets the interval over which a change in the equilibrium is sought, and a dwell time  $T_d$ , which sets how long that change must persist before it is declared significant. Both depend on the response time of the conditioning filters and the settling time of the slowest PLL the plant operator may employ, and the dwell time should exceed the duration of the transients expected under both normal and adverse grid conditions, while the magnitude and clarity of the equilibrium offset can be optimized by tuning  $\omega_N$ . Accordingly, there is a trade-off: a small dwell time yields early detection but may register transients as tampering, whereas a larger dwell time delays detection but ensures no false positives occur. However, since the impact of this attack manifests only during significant transients, which are unlikely to occur within even a few seconds of the attack, a more conservative dwell time is generally preferable. Moreover, since the equilibrium points are functions only of the PLL gains and the grid frequency, the latter being practically constant over the timescale of the attack, a sustained equilibrium offset can arise solely from a gain change, thereby precluding false positives.

The algorithmic equivalent of the detection pipeline is presented in Algorithm 1. At every sample, the equilibrium vector  $\eta^s$  is compared with its value one observation window  $T_w$  earlier, and whenever a component of the resulting difference persistently exceeds its corresponding threshold,  $\epsilon_{\Delta\theta}$  or  $\epsilon_z$ , for longer than the dwell time  $T_d$ , the associated flag,  $\sigma_{\Delta\theta}$  or  $\sigma_z$ , is set to one, signifying a gain-change event. The thresholds and timing parameters must be selected relative to the noise level, the filtering parameters, and, most importantly, the steady-state levels of the equilibrium points. To accommodate possible gain adaptation, the observation window can be referenced to the slowest PLL the operator or adaptation algorithm may employ and conservatively set to a few seconds. Even such a conservative window detects the tampering well before it can escalate into harmful operation, whereas a less conservative design, tuned to the neighborhood of the actual PLL bandwidth in use, reduces the detection delay to a few hundred

#### Algorithm 1 RSU-Based Detection of PLL Gain Tampering

- 1: **Constants:**  $\omega_N$ , sampling time  $T_s$
- 2: **Input:** equilibrium vector  $\eta^s = (\Delta\theta^s, z^s)$ ; window  $T_w$ ; dwell time  $T_d$ ; thresholds  $\epsilon = (\epsilon_{\Delta\theta}, \epsilon_z)$
- 3: **Output:** flags  $\sigma = (\sigma_{\Delta\theta}, \sigma_z)$
- 4:  $f \leftarrow 0$ ,  $\sigma \leftarrow 0$ ,  $w \leftarrow T_w/T_s$
- 5: Filter  $\eta^s$  through the conditioning low-pass filter
- 6: **for** each sample  $n$  **do**
- 7:  $\Delta\eta \leftarrow |\eta^s(n) - \eta^s(n-w)|$  ▷ element-wise windowed difference
- 8: **for** each component  $i \in \{\Delta\theta, z\}$  **do**
- 9: **if**  $\Delta\eta_i > \epsilon_i$  **then**
- 10:  $f_i \leftarrow f_i + T_s$  ▷ accumulate dwell
- 11: **else**
- 12:  $f_i \leftarrow 0$  ▷ reset on transient
- 13: **end if**
- 14: **if**  $f_i \geq T_d$  **then**
- 15:  $\sigma_i \leftarrow 1$  ▷ gain-change event on component  $i$
- 16: **end if**
- 17: **end for**
- 18: **end for**

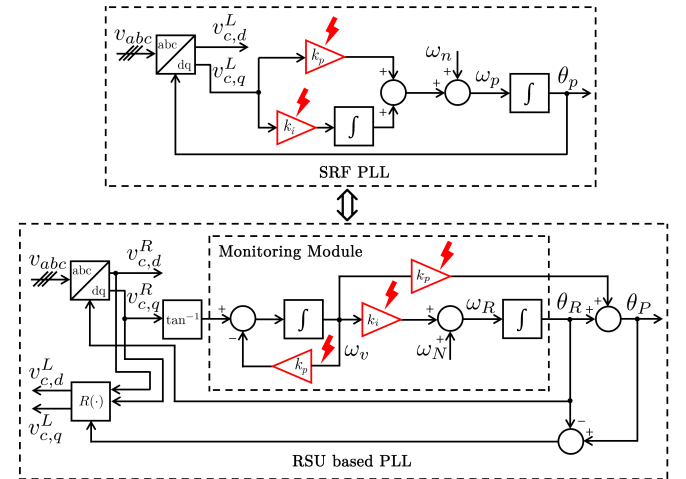


Fig. 9: Proposed RSU-based PLL.

milliseconds. A malicious gain tampering may go unnoticed only when it produces an equilibrium change smaller than  $\epsilon_{\Delta\theta}$  or  $\epsilon_z$ , which occurs for marginal gain changes that in turn have a negligible impact on the PLL response, thereby ensuring that any tampering capable of degrading system performance is reliably detected.

To demonstrate how the proposed design does not disturb the transient performance of the conventional SRF PLL, the transfer function between  $\theta_P$  and  $\theta_g = \int \omega_g dt$  is formulated:

$$\frac{\theta_P}{\theta_g} = \frac{k_p s + k_i}{s^2 + k_p s + k_i}, \quad (44)$$

which obviously coincides with that of the SRF PLL, contrary to the alternative synchronization designs presented in [18], [19] that fundamentally alter the dynamics of the PLL. Therefore, the proposed and conventional SRF PLLs are small-signal equivalent and exhibit identical small-signal responses. Consequently, since they share a common input-output relation, any small-signal stability analysis carried out for the SRF PLL yields exactly the same results for the proposed surrogate

RSU-based PLL, including that of Section V. Moreover, given its low computational complexity, comprising the same number of states as the SRF PLL, and owing to this equivalence, the proposed structure can be readily integrated into existing GFL configurations, augmenting them with the ability to detect malicious gain tampering while fully preserving the tracking performance of the original PLL.

Another important advantage of the proposed scheme over [18] is that it can be readily adopted to any GFL control structure. Specifically, the controller in Fig. 2 requires accurate current decoupling to actively isolate the active and reactive power responses. This decoupling relies on an explicit formulation of the Park transformation rotating frequency, which is not available in [18], [19]. In contrast, for the proposed approach, the rotating frequency is given by:

$$\begin{aligned}\omega_P &= \omega_N + k_i z + k_p \dot{z}, \\ &= \omega_N + k_i z + k_p (\arctan v_{c,q}^R - k_p z).\end{aligned}\quad (45)$$

The proposed surrogate PLL can also be combined with the same quadratic signal generator (QSG)-based positive- and negative-sequence separation employed by a conventional SRF PLL [26], thereby enabling operation under unbalanced networks and dominant low-order harmonics, such as the 5<sup>th</sup> and 7<sup>th</sup>. The integration is a drop-in modification: the QSG sequence-separation stage is retained unchanged, and only the synchronization-loop equations, i.e. (22)-(23) are replaced by those of the RSU-based PLL in (40)-(41), with the frequency-adaptive resonators driven by the RSU output frequency (45). The small-signal equivalence between the two PLLs is preserved under this implementation, since their small-signal input-output relation is identical and the QSGs derive the filtered positive- and negative-sequence signals from the output frequency of each PLL, which, as proven above, are small-signal equivalent. The same conditioned sequence signals are therefore presented to both loops, leaving the equivalence intact.

#### A. Extension of the Detection Scheme to Multi-Layer IBR Control

Extending the proposed detection scheme to other IBR controllers is straightforward and relies on the uniqueness of the equilibrium points of the power system. Unlike PLLs, these controllers can be suitably modified such that any change in a controller gain produces a unique effect on the equilibrium point of the controllers' internal states. Without loss of generality, the following alternative PI controller is considered:

$$u = k_p (ax_r - x) + k_i \int (x_r - x), \quad (46)$$

where  $x$  and  $x_r$  denote the measured and reference states, respectively, and  $a \in (0, 1)$  is a constant introduced to mask the controller's equilibrium point.

Implementing this PI controller in place of a conventional one yields unique, nonlinear equilibrium-point mappings of the form:

$$\mathbf{0} = \underbrace{\mathbf{g}(k_p, k_i, x_r)}_{\text{Nonlinear function}}, \quad (47)$$

TABLE II: Experimental testbench parameters.

| Parameter                       | GFL Inverter | GFM Inverter           |
|---------------------------------|--------------|------------------------|
| Power Rating                    | 0.1 MVA      | 1 MVA                  |
| Frequency Droop                 | 5%           | 2%                     |
| Voltage Droop                   | 5%           | 5%                     |
| <b>Common System Parameters</b> |              |                        |
| Base Load                       |              | 0.3 MVA                |
| Line                            |              | 0.01 $\Omega$ , 0.2 mH |
| Switching Frequency             |              | 15 kHz                 |
| Nominal Line Voltage            |              | 400 V                  |

which enable the detection of controller gain tampering by recovering the controller gains from the solutions of  $\mathbf{g} = \mathbf{0}$ . Such gain recovery is feasible only for the plant operator, who, analogous to the case of the PLL and the parameter  $\omega_N$ , is the sole entity with knowledge of the controller structure and the parameters  $\mathbf{a}$ . This property significantly complicates attempts by an adversary to mask equilibrium-point alterations, thereby enhancing the resilience of the SRF PLL-based control architecture against attacks targeting critical IBR infrastructure.

## VII. EXPERIMENTAL RESULTS

This section presents an experimental validation of the proposed detection scheme under realistic operating conditions using a CHIL setup, depicted in Fig. 10. Two scenarios are examined, under balanced and under unbalanced, distorted grid conditions, respectively. Each scenario establishes the small-signal equivalence between the RSU-based and conventional SRF PLLs, and further demonstrates the real-time applicability of the proposed PLL and its ability to detect the adopted threat model. The attack is applied to a GFL IBR, whose firing pulses are generated by a controller implemented on a Speedgoat Baseline real-time target, while the cyber-physical system is emulated in the Typhoon HIL 404 environment.

### A. Operation and Attack Detection under Balanced Grid Conditions

The system shown in Fig. 3 is considered, where a GFL converter is connected to a grid emulated by a GFM converter. The critical system and controller parameters are summarized in Table II. The synchronization PLL bandwidth is set to 20 Hz and the frequency-estimation bandwidth to 2.5 Hz, with  $\zeta = 1/\sqrt{2}$  and  $f_N = 49.9$  Hz  $= \omega_N/2\pi$ . Furthermore, the detection algorithm parameters are set to  $\epsilon_{\Delta\theta} = 2 \times 10^{-2}$ ,  $\epsilon_z = 5 \times 10^{-4}$ , and  $T_d = T_w = 0.8$  s. The two objectives are to demonstrate that the proposed PLL implementation behaves identically to the conventional SRF PLL, and to verify the effectiveness of the detection framework. All presented signals are filtered using a low-pass filter with a 100 Hz cut-off frequency.

Fig. 11 illustrates the system response of the grid-supporting IBR employing the proposed and conventional PLLs under various contingencies, including reference power and load changes. The theoretical analysis is validated through the active and reactive power responses and the frequency estimation shown in Figs. 11a-11c. As expected, the GFL converter smoothly tracks the reference setpoint while the system frequency remains within the deadband, thereby nullifying grid-support services. When the frequency exits the  $\pm 0.1$  Hz band at  $t = 7.5$  s due to an abrupt load increase (15 kW, 45 kVar),



Fig. 10: Experimental benchmark.

the power output is immediately adjusted to accommodate the new operating conditions.

To demonstrate the attack-detection capability of the proposed PLL, the adopted threat model is applied by actively altering the PLL gains of the frequency estimator. For comparison, the detector introduced in [18], which enables detection based on the phase estimation error, is also considered. In the first scenario, the adversary sets  $k_i^a = k_i^n/3$  at  $t = 5$  s. The results are shown in Fig. 12. Immediately after the attack, all system states are perturbed, including the active power and estimated frequency depicted in Figs. 12a and 12d, which subsequently converge to the same equilibrium point. However, the equilibrium points of the internal phase error (Fig. 12b) and  $z$  (Fig. 12c) are significantly shifted, indicating a gain manipulation. This is also observed in Fig. 12e, where both flags switch to one approximately 0.8 s after the inception of the attack, as dictated by  $T_d$ . In contrast, the detector of [18], based on the phase estimation error of the SRF PLL, fails to detect the abnormality, and the system remains undisturbed during the gain change. This behavior is consistent across additional attack scenarios involving more complex gain-tampering patterns.

The attack is also detected when both or only the proportional gain is modified. The system responses for  $k_p^a = k_p^n/2$ ,  $k_i^a = 3k_i^n$  and  $k_p^a = 2k_p^n$ , are shown in Figs. 13 and 14, respectively. In both cases, the frequency estimator continues to track the grid frequency (Figs. 13a and 14a), while the active power output (Figs. 13d and 14d) is significantly disturbed before converging to the original equilibrium. In contrast, the equilibrium points of  $z$  (Figs. 13c and 14c) and the phase error (Figs. 13b and 14b) are notably shifted, enabling attack detection. As in the previous scenario, because the integral action remains active, the detector of [18] cannot identify the attack, which remains completely stealthy from its perspective. In practice, the minimum detection delay is upper-bounded by the sum of the settling times of the conditioning filter ( $\approx 0.04$  s) and the PLL ( $\approx 0.34$  s), since the two stages respond in series; this yields a conservative estimate of roughly 0.38 s, the true combined response being somewhat faster. The dwell time  $T_d$  can therefore be set from this floor, for fastest response, up to a few seconds; here it is set higher to cover the settling of any PLL configuration the operator may employ.

#### B. Performance under Unbalanced and Distorted Grid Conditions

This section assesses the proposed detection mechanism under unbalanced and distorted grid conditions. A stiff grid bus is considered, with phases  $a$ ,  $b$ , and  $c$  set to  $0.98 \angle +0.5^\circ$  pu,  $1.05 \angle -1.5^\circ$  pu, and  $0.96 \angle -0.5^\circ$  pu, respectively, introducing a non-negligible negative-sequence component in addition

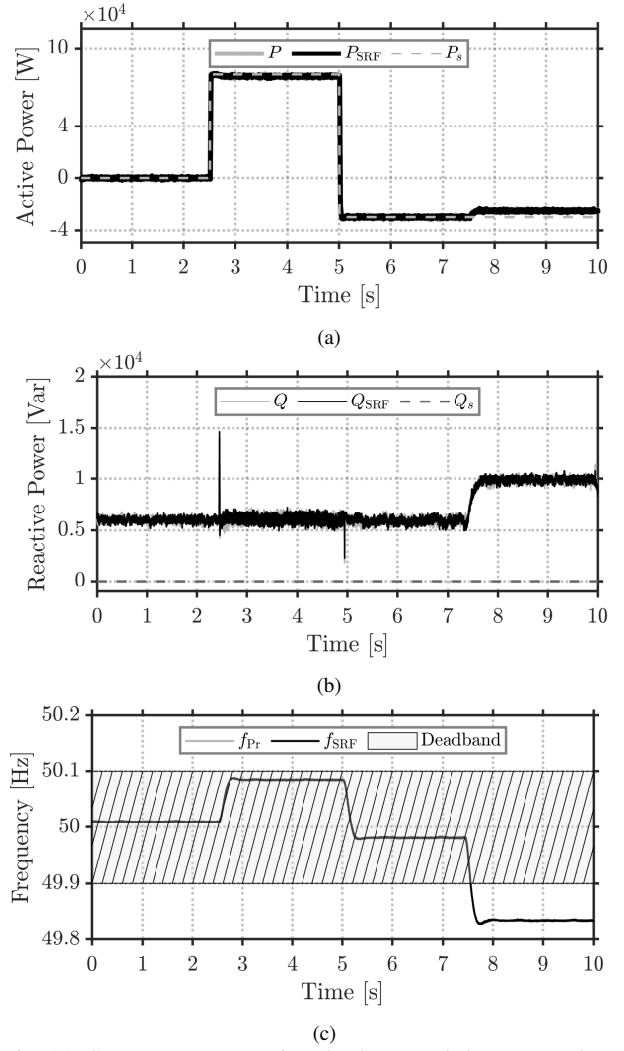


Fig. 11: System response using the SRF- and the proposed PLL.

to the nominal positive-sequence voltage. To reject the resulting distortion, the PLLs are equipped with decoupling QSG networks [26], and the synchronization PLL bandwidth is set to 25 Hz given the grid stiffness. Furthermore, the deadband is tightened from  $\pm 0.1$  Hz to  $\pm 0.05$  Hz to stress-test the detection scheme under the resulting increase in primary-controller activity and noise sensitivity. In this benchmark, a single PLL performs both synchronization and frequency estimation, as in [17], with  $\omega_o$  obtained by low-pass filtering the PLL frequency at 5 Hz. Here,  $f_N = 46$  Hz  $= \omega_N/2\pi$ , increasing the difference  $|\omega_g - \omega_N|$  and thereby enlarging the equilibrium offsets and the sensitivity to the PLL gains, which facilitates detection. Now, the detection parameters are set to  $\epsilon_{\Delta\theta} = 5 \times 10^{-3}$ ,  $\epsilon_z = 1 \times 10^{-4}$ , and  $T_d = T_w = 1$  s. Furthermore, the switching frequency is now reduced to 10 kHz to demonstrate that the proposed detection scheme remains effective even under lower sampling rates, underscoring its real-time applicability and industrial integration. Two scenarios are examined. The first establishes that the proposed and conventional SRF PLLs, both employing QSG networks, remain equivalent and robust under these conditions. The second verifies that the detection scheme reliably exposes PLL gain tampering despite the negative-sequence component and harmonic distortion.

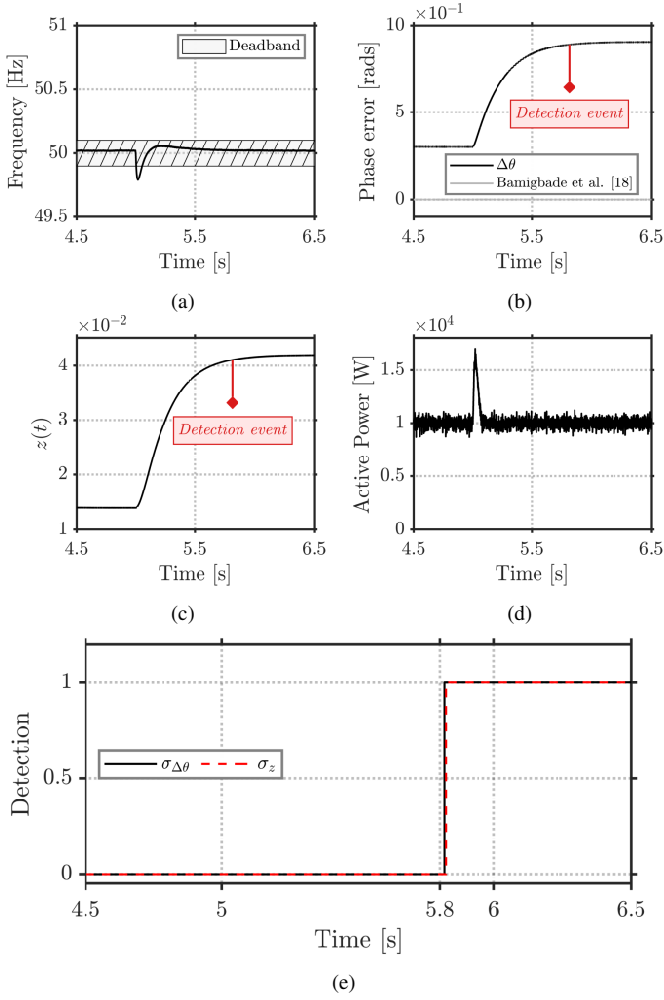


Fig. 12: System response after integral-only gain tampering.

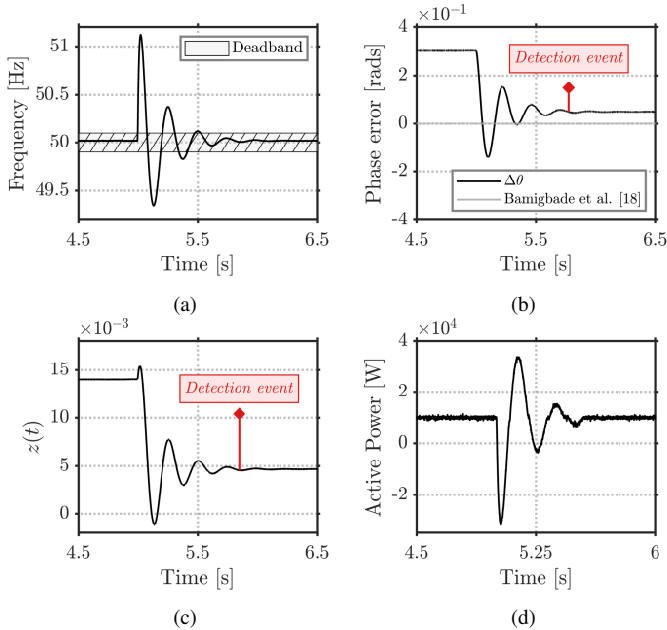


Fig. 13: System response after proportional-integral gain tampering.

The first scenario applies a sequence of disturbances. The grid frequency is stepped to 50.1 Hz at  $t = 2$  s and to 49.9 Hz at  $t = 4$  s. A voltage sag is then imposed at  $t = 6$  s,

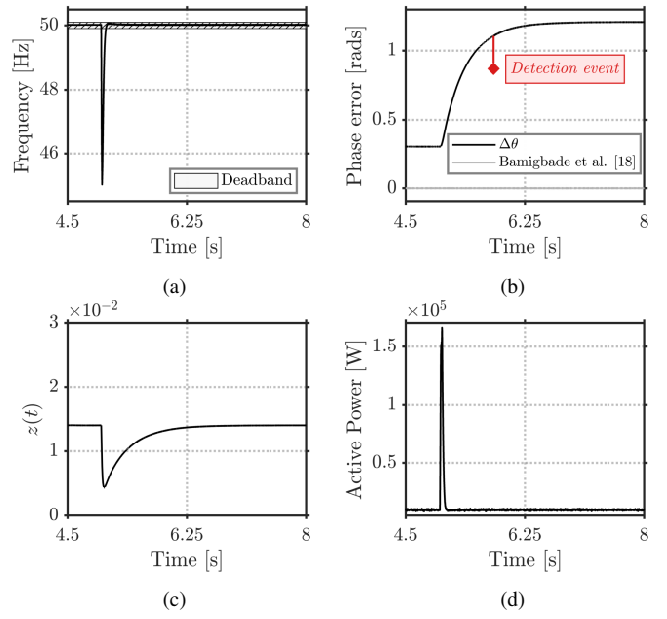


Fig. 14: System response after proportional gain tampering.

during which the magnitude of each phase drops to 70 %, and is cleared at  $t = 8$  s. Finally, at  $t = 10$  s, 5<sup>th</sup> and 7<sup>th</sup> harmonics of magnitude 0.06 pu and 0.09 pu are injected, respectively. The system response is shown in Fig. 15. As in the balanced-grid case, the two implementations perform practically identically. The active power (Fig. 15a) smoothly tracks the outer-loop frequency-support commands following the frequency steps, while the reactive power (Fig. 15b) adheres to the primary controller, saturating at 14 kVar during the sag at  $t = 6$  s and recovering to its pre-fault value once the fault clears. The frequency estimates (Fig. 15c) remain virtually identical even through transients, as the zoomed-in plots confirm. Finally, the resilience to higher-order harmonics is evident at  $t = 10$  s, where both PLLs continue to operate correctly despite the increased harmonic content in the measured signals.

The effectiveness of the detection scheme is demonstrated in the second scenario, illustrated in Fig. 16, under the same unbalanced grid with constant injection of the 5<sup>th</sup> and 7<sup>th</sup> harmonics of the previous test case. A voltage sag of 0.7 pu occurs at  $t = 4$  s and is cleared at  $t = 4.5$  s. The adversary exploits this disturbance as cover, targeting the synchronization PLL and setting  $k_p^a = 0.8k_p^n$  at  $t = 6$  s, shortly after the sag clears. Despite the concurrent disturbances, the proposed PLL operates nominally and continues to track the grid frequency, as shown in Fig. 16a. Since the PLL equilibrium points depend only on the grid frequency and on the PLL gains, neither the voltage sag nor the harmonic distortion can mask the effect of the proportional-gain tampering. The tampering appears as a transient in the power output (Fig. 16d) and in the internal state  $z$  (Fig. 16c), which returns to its pre-attack equilibrium as expected from (43) since  $z^s$  is independent of  $k_p$ , and, more importantly, as a persistent offset in the equilibrium of the internal phase error (Fig. 16b), which the plant operator can observe as evidence that the gains have been altered. Evidently, as shown in Fig. 16e, after approximately one

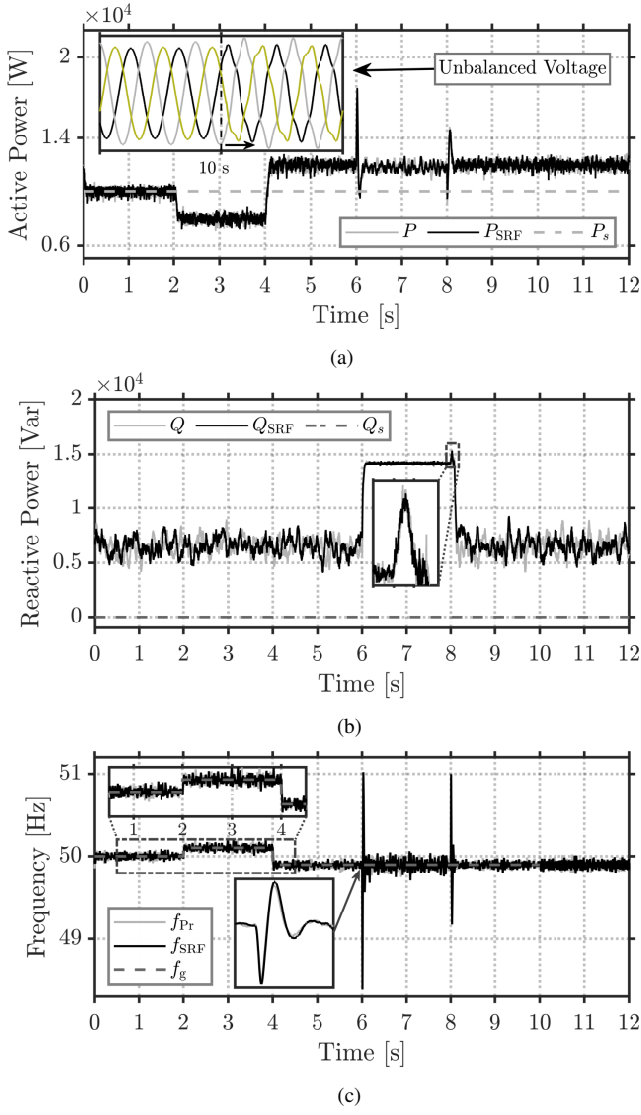


Fig. 15: System response using the SRF and the proposed PLLs under unbalanced and distorted grid voltage.

second, dictated by  $T_d$ , the phase-error detector is activated, indicating the tampering, while, as in the balanced-grid case, the detector of [18] again remains unperturbed and unaware of the compromise. In this case, the minimum detection delay comprises  $\approx 0.04$  s for the conditioning filter and  $\approx 0.04$  s for the PLL settling time, giving a lower bound of  $\approx 0.08$  s. As before, this delay is small enough to leave the attack little opportunity to manifest before the operator is alerted; here, however, similarly to the previous scenario,  $T_d$  is set higher, to 1 s, to cover the settling times of all plausible gain combinations and potential transients under normal or adverse operating conditions.

## VIII. CONCLUSION

This paper demonstrates how an adversary who overcomes local system defenses can manipulate critical control gains of PLLs responsible for synchronizing and controlling grid-supporting GFL IBRs, thereby limiting ancillary services and, in severe cases, destabilizing the underlying IBR infrastructure. To characterize this threat, an explicit analysis accounting for all cyber-physical and control aspects of grid-connected

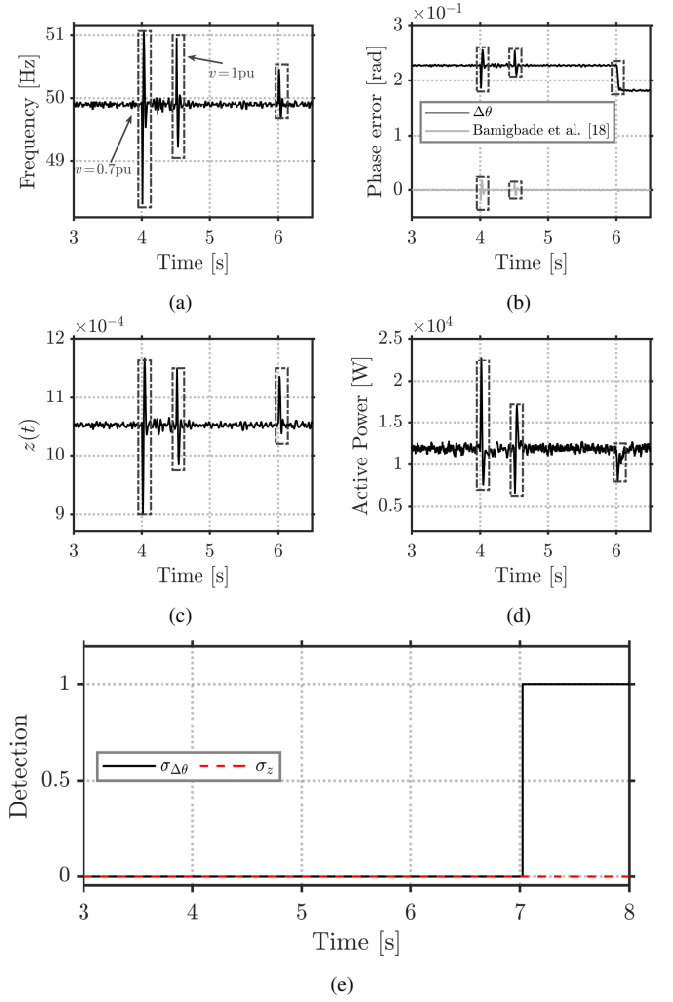


Fig. 16: System response after proportional gain tampering under unbalanced and distorted grid voltage.

converters, including line and grid dynamics, was conducted, revealing the road-map an adversary could follow to compromise IBR resources stealthily. The analysis further shows that, while the additional frequency-support loop improves the stability margins and overall small-signal behavior of the converters, poor tuning can provoke severe interactions between control layers and even instability.

To counter this threat, an alternative PLL implementation based on the RSU was introduced. Unlike prior approaches, the proposed scheme preserves the performance of the SRF PLL, has low computational complexity and is therefore readily applicable to existing GFL infrastructure, and, most importantly, detects both proportional and integral gain tampering by monitoring the internal equilibrium points, which shift whenever either gain is altered. Its effectiveness was validated through extensive CHIL benchmarks under normal, unbalanced, and distorted operating conditions, confirming reliable detection of the otherwise stealthy attack.

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